

Microsoft Azure AI-900

Practice Question Bank

10 Exam-Style Questions with Full Explanations

Score yourself: 8-10 = Exam Ready | 5-7 = Review Notes | <5 = Study More

Covers all 6 exam domains • Multiple-choice format • Detailed answer explanations

Domain	Questions	Difficulty
AI Concepts	Q1	Easy
Machine Learning	Q2, Q7	Medium
Computer Vision	Q3, Q8	Medium
NLP	Q4	Easy
Generative AI	Q5, Q9	Medium
Responsible AI	Q6, Q10	Medium

Questions 1–5

Domains: AI Concepts · ML · Computer Vision · NLP · Generative AI

1

AI Concepts

A company wants to automatically sort customer emails into categories like 'Billing', 'Technical Support', and 'General Enquiry'. Which type of AI workload best describes this scenario?

A. Anomaly detection

✓ B. Text classification

C. Object detection

D. Speech synthesis

Correct Answer: B. Text Classification

Text classification assigns a predefined label to a piece of text. Sorting emails into named categories is a classic classification task. Anomaly detection finds unusual patterns; object detection works on images; speech synthesis converts text to audio — none of these fit here.

2

Machine Learning

You train a machine learning model on a dataset of house prices. On training data it achieves 98% accuracy, but on new unseen data it only achieves 61%. What problem does this describe?

A. Underfitting

B. Data leakage

✓ C. Overfitting

D. Class imbalance

Correct Answer: C. Overfitting

Overfitting occurs when a model learns the training data too well — including its noise and outliers — and therefore fails to generalise to new data. Underfitting would give poor accuracy on BOTH sets. Data leakage would inflate test accuracy, not decrease it.

3

Computer Vision

A retailer wants to count the number of people currently in each aisle of their store using ceiling-mounted cameras. Which Azure AI service is most appropriate?

A. Azure AI Custom Vision

B. Azure AI Face

✓ **C. Azure AI Vision — Spatial Analysis**

D. Azure AI Document Intelligence

Correct Answer: C. Azure AI Vision — Spatial Analysis

Spatial Analysis (part of Azure AI Vision) is specifically designed to count people, track movement, and analyse zones in video streams. Custom Vision trains custom image classifiers. Face detects/identifies faces but isn't optimised for crowd counting in video. Document Intelligence extracts data from documents.

4

NLP

A healthcare provider wants to scan thousands of patient feedback forms and automatically detect whether each comment is positive, negative, or neutral. Which Azure AI Language feature should they use?

A. Named Entity Recognition (NER)

✓ **B. Sentiment Analysis**

C. Key Phrase Extraction

D. Language Detection

Correct Answer: B. Sentiment Analysis

Sentiment Analysis classifies text as positive, negative, or neutral (and can provide a confidence score). NER extracts entities like names and dates. Key Phrase Extraction identifies important phrases but not sentiment. Language Detection identifies which language the text is written in.

5

Generative AI

A developer is building a customer service bot that answers questions using the company's internal knowledge base. Which technique allows the large language model to reference this private data without retraining?

A. Fine-tuning

B. Prompt injection

✓ **C. Retrieval-Augmented Generation (RAG)**

D. Transfer learning

Correct Answer: C. Retrieval-Augmented Generation (RAG)

RAG retrieves relevant documents from an external knowledge base at query time and includes them in the prompt context, allowing the LLM to answer using private data without retraining. Fine-tuning updates model weights with new data (expensive). Prompt injection is a security attack, not a technique.

Questions 6–10

Domains: Responsible AI · ML · Computer Vision · Generative AI · Responsible AI

6

Responsible AI

An AI hiring tool is found to recommend male candidates at a significantly higher rate than equally qualified female candidates. Which Microsoft Responsible AI principle is being violated?

- A. Reliability & Safety
- B. Transparency
- ✓ C. Fairness
- D. Privacy & Security

Correct Answer: C. Fairness

Fairness requires that AI systems treat all individuals equitably and do not discriminate based on characteristics such as gender, race, or age. This scenario is a textbook example of algorithmic bias violating the Fairness principle. Microsoft recommends using tools like Fairlearn to assess and mitigate such bias.

7

Machine Learning

A data scientist evaluates her binary classifier and finds: Precision = 0.95, Recall = 0.40. What does this tell her about the model's behaviour?

- ✓ A. The model misses many true positives but rarely gives false alarms
- B. The model catches most positives but has many false positives
- C. The model performs equally well on both classes
- D. The model is overfitting to the negative class

Correct Answer: A. The model misses many true positives but rarely gives false alarms

High Precision (0.95) means: when the model predicts positive, it is almost always right — very few false positives. Low Recall (0.40) means: the model only finds 40% of actual positives — it misses 60%. This pattern is typical of a conservative model with a high decision threshold.

8

Computer Vision

Which Azure service would you use to extract structured data — such as invoice number, date, and line items — from scanned PDF invoices?

- A. Azure AI Vision OCR
- ✓ B. Azure AI Document Intelligence

C. Azure AI Custom Vision

D. Azure AI Search

Correct Answer: B. Azure AI Document Intelligence

Azure AI Document Intelligence (formerly Form Recognizer) is purpose-built to extract structured fields and tables from forms, invoices, receipts and contracts. Plain OCR only extracts raw text without understanding document structure. Custom Vision handles image classification/detection, not document extraction.

9

Generative AI

Which of the following best describes the role of 'temperature' when generating text with a large language model?

A. It controls the maximum number of tokens the model can output

B. It determines how much of the training data the model uses

✓ **C. It controls the randomness and creativity of the model's output**

D. It sets the confidence threshold for the model's predictions

Correct Answer: C. It controls the randomness and creativity of the output

Temperature is a sampling parameter: a low value (e.g. 0.1) makes output predictable and deterministic; a high value (e.g. 1.5) makes it more varied and creative. Max tokens controls output length. Temperature has no bearing on training data or confidence thresholds.

10

Responsible AI

A bank deploys an AI model that denies loan applications. Customers are told 'the AI decided' but receive no explanation of the factors involved. Which TWO Responsible AI principles are most clearly being violated?

A. Fairness and Reliability

✓ **B. Transparency and Accountability**

C. Privacy and Inclusiveness

D. Reliability and Privacy

Correct Answer: B. Transparency and Accountability

Transparency requires that AI decisions are understandable and explainable to those affected. Accountability means humans (not 'the AI') are responsible for decisions. Saying 'the AI decided' with no explanation violates both principles. In regulated industries like finance, explainability is also a legal requirement.

Score Card

Tally your correct answers and see where you stand

Score	Result	Recommendation
9 – 10	Exam Ready ■	Book your exam — you're well prepared!
7 – 8	Almost There ■	Review any wrong answers, then sit the exam
5 – 6	More Revision Needed ■	Re-read the study guide domains you missed
< 5	Keep Studying ■	Work through the full study guide again

Domain Breakdown — Where Did You Drop Marks?

Domain	Questions	Your Score
AI Concepts	Q1	/ 1
Machine Learning	Q2, Q7	/ 2
Computer Vision	Q3, Q8	/ 2
NLP	Q4	/ 1
Generative AI	Q5, Q9	/ 2
Responsible AI	Q6, Q10	/ 2